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Chapter · November 2015

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Order stars and stability domains

Ernst Hairer[1], Gerhard Wanner[1]

Section de mathématiques, 2-4 rue du Lièvre, Université de Genève, CH-1211 Genève 4, Switzerland. E-mail: Ernst.Hairer@unige.ch, Gerhard.Wanner@unige.ch

Throughout the 20th century, and before, scientists were struggling to find numerical methods for initial value problems satisfying the requirements

- High computational speed,
- High precision,
- High stability.

Order stars turn these analytical properties into geometrical quantities. They are thus not only attractive for the importance of their theoretical results, but also for aesthetics and beauty.

Stability function

While the first two of the above requirements are more or less trivial claims, the importance of the third one became clear, to many researchers, only after several numerical disasters. Stability theory for numerical methods started with Courant–Friedrichs–Lewy [3]; the classical papers Dahlquist [4] and Guillou-Lago [7] initiated stability theory for stiff equations. The principal tool is *Dahlquist's test equation*

$$\dot{y} = \lambda y$$

which models the behavior of stiff systems in the neighborhood of a stationary point. Here, λ represents one of the (possibly complex) eigenvalues of the Jacobian matrix of the vector field. The equation is stable "in the sense of Lyapunov" if $\Re\lambda \leq 0$. If a one-step method $y_{n+1} = \Phi_h(y_n)$ is applied to this equation, one usually obtains a relation

$$y_{n+1} = R(z)\, y_n \quad \text{i.e.} \quad y_n = (R(z))^n y_0 \quad \text{where} \quad z = h\lambda$$

with a function $R(z)$ which is called *stability function* of the method. We see that $|R(z)| \leq 1$ is necessary for stability, and we call the set

$$S := \{z \in \mathbb{C}\ ;\ |R(z)| \leq 1\} \tag{1}$$

its *stability domain*. The criterion for stability requires that all eigenvalues λ, multiplied by the step size h, must lie in S. For hyperbolic PDE's this is known as the CFL-condition.

Examples of stability functions and stability domains are presented in Fig. 1 for the explicit Euler method (see entry "one-step methods"), Runge's method

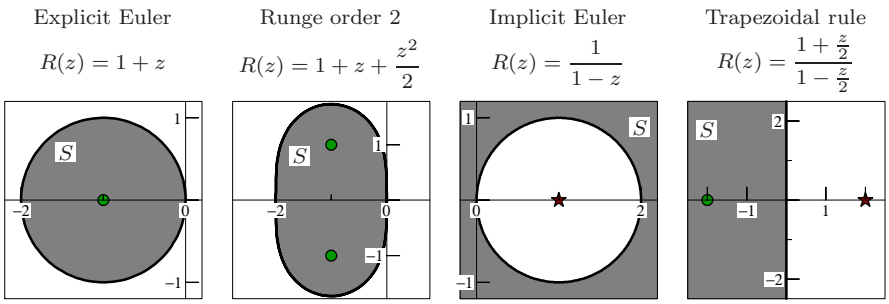

Fig. 1. Stability functions and stability domains for the oldest one-step methods.

of order 2 (see entry "Runge–Kutta methods"), the implicit Euler method and the trapezoidal rule.

We observe that for *explicit* methods, with k function evaluations per step, $R(z)$ is a polynomial of degree k, which leads to the existence of k zeros (marked by ●) lying inside of S. The corresponding stability domain is a bounded set, so that severe step size restrictions can occur for stiff equations. On the contrary, *implicit* stages lead to poles (marked by ★), so that $R(z)$ becomes rational which allows both methods to be *A-stable*, i.e. the stability domain covers the entire left half plane of $\mathbb{C}$. Here, for stable problems, there are no stability restrictions, independent of how stiff the equation is.

Implicit Runge–Kutta methods. It was discovered by Ehle [5] that the stability functions of high order implicit Runge–Kutta methods are *Padé* approximations to the exponential function

$$R_{k,l}(z) = \frac{P_{k,l}(z)}{Q_{k,l}(z)}$$

where

$$P_{k,l}(z) = \sum_{j=0}^{k} \binom{k}{j} \frac{(k+l-j)!}{(k+l)!} z^j \qquad \text{and} \qquad Q_{k,l}(z) = P_{l,k}(-z).$$

For example, the stability functions in Fig. 1 are, from left to right, $R_{1,0}(z)$, $R_{2,0}(z)$, $R_{0,1}(z)$, and $R_{1,1}(z)$. Butcher's implicit Gauss methods are on the diagonal of the Padé table ($k = l = s$) and Radau IIA methods are on the first sub-diagonal ($k = s-1$, $l = s$; for more details see Table 3 in entry *Runge–Kutta methods*). Ehle proved that both types of methods are A-stable for all $s \geq 1$ (see e.g. Fig. 1 in entry "Radau methods"). Ehle also established the *Conjecture* that only the diagonal Padé and the first two sub-diagonals are A-stable.

Order stars

The attempts to understand Ehle's results and prove Ehle's conjecture, which lasted years, led finally to the idea of the order stars [12]. The motivation comes from a careful inspection of the stability function, for example the function $|R_{2,3}(z)|$ (for Radau5) drawn in Fig. 2 to the left. We observe that the stability

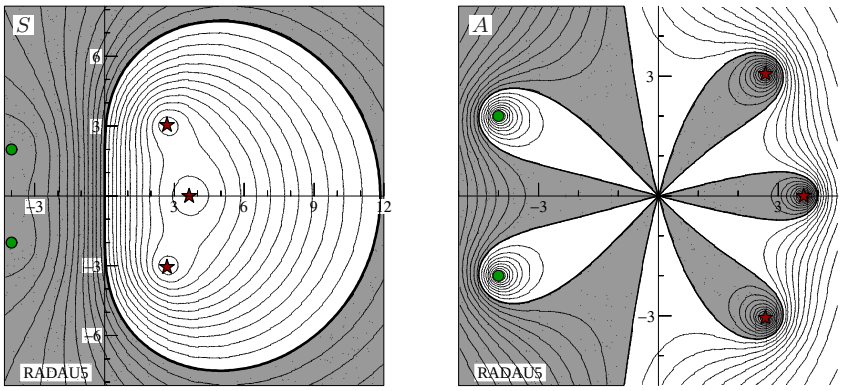

Fig. 2. Stability domain (left) and order star (right) for $R_{2,3}$ (Radau5).

domain is the outside of a potato-like curve which surrounds the three poles. Looking at the level curves in the neighborhood of the origin, we see that they mimic the parallel level curves of the exponential function. Since $R(z)$ is a high-order approximation of e^z, we expect more insight if we compare the function $|R(z)|$, not to 1 as in (1), but to $|\mathrm{e}^z|$. This motivates to consider the set

$$A := \{z \in \mathbb{C} \ ; \ |R(z)| > |\mathrm{e}^z|\} \tag{2}$$

(see Fig. 2 to the right), which we call *order star*. It transforms numerical quantities into geometric properties as follows:

- *Computational speed:* This is determined by the number of stages, which the method uses at each step. Explicit function evaluations transform into zeros ●, each implicit stage transforms into a pole ★. The more zeros or the more poles are present, the more work requires the method.
- *High precision:* This is characterized by the *order* p of the method and the *error constant* C, which (for the test equation $\dot{y} = \lambda y$) are given by

$$\mathrm{e}^z - R(z) = Cz^{p+1} + \mathcal{O}(z^{p+2}) \ .$$

  As a consequence, the set A is star-like with $p+1$ "fingers" of regular width coming out from the origin. The error constant determines the color of these fingers.
- *A-stability:* Along the imaginary axis, where $|\mathrm{e}^z| = 1$, the order star is complementary to the stability domain S. So, if we know that the order star does not contain part of the imaginary axis and if, in addition, all poles stay in the right half plane, we have A-stability by the maximum principle. Inversely, if the order star covers a part of the imaginary axis, we have no A-stability.

Some elementary complex analysis (the argument principle) allows the conclusion that all bounded black fingers must contain a pole, and that every bounded subset of A, whose boundary returns j times to the origin, must contain j poles (see e.g. "Lemma 4.5" of [8]). Ehle's results and Ehle's conjecture can now be directly seen from the right picture of Fig. 2, because only for $k \le l \le k+2$ the imaginary axis can pass between the white and black fingers without touching the order star (for details see Theorems 4.6, 4.7, 4.8, 4.9, 4.10 and 4.11 of [8]).

Multistep methods

We choose as example the two-step explicit Adams method

$$y_{n+2} = y_{n+1} + h\Big(\frac{3}{2}f_{n+1} - \frac{1}{2}f_n\Big)\,.$$

For Dahlquist's test equation we obtain $y_{n+2} - (1+\frac{3}{2}z)y_{n+1} + \frac{1}{2}zy_n = 0$, i.e. $y_n = c_1R_1^n + c_2R_2^n$ where R_1 and R_2 are the roots of the characteristic equation

$$R^2 - \Big(1+\frac{3}{2}z\Big)R + \frac{1}{2}z = 0\,.$$

For $z = 0$ this equation becomes $R^2 - R = 0$ with roots $R_1 = 1$ (the principal root) and $R_2 = 0$ (the parasitic root). Both roots continue as complex functions of z and we have stability where *both* roots are bounded by 1. This determines the stability domain (see the first two pictures of Fig. 3). At certain "branching points" ⧗ these roots merge and we observe a discontinuity in their neighbourhood. Thus we place the two roots one above the other (third picture of Fig. 3) and obtain *one* analytic function $R(z)$ on the *Riemann surface* M.

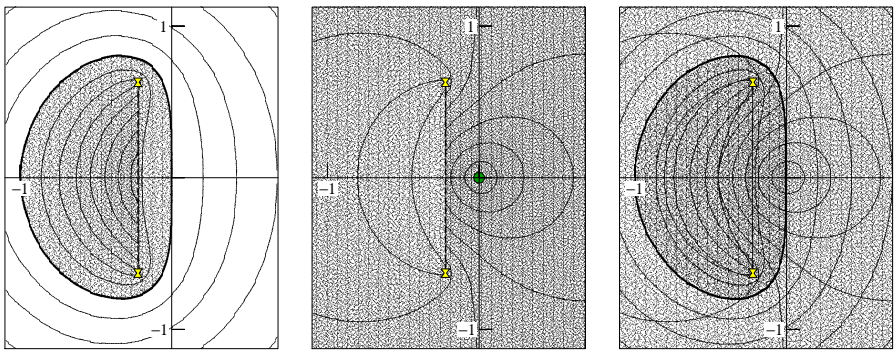

Fig. 3. Stability domain for Adams2; principal root $|R_1(z)|$ (left); parasitic root $|R_2(z)|$ (middle); both roots on the Riemann surface (right).

Order stars for multistep methods. The definition (2) of order stars carries over to Riemann surfaces without changes. There will be a star on the principal sheet with $p+1$ sectors, an explicit stage leads to a zero on one of the sheets (see Fig. 4), each implicit stage gives rise to a pole on one of the sheets and for A-stability the order star must stay away from the imaginary axis on *all* sheets.

Daniel–Moore conjecture. An A-stable multistep method with k implicit stages has order $p \le 2k$. The proof of this statement is illustrated in Fig. 5. This order barrier was originally conjectured for multistep-Taylor methods in 1970 and became a theorem in 1978. The special case for linear multistep methods, where $k = 1$ and hence $p \le 2$, is *Dahlquist's second order barrier* from 1963.

Jeltsch–Nevanlinna theorem. During the first half of the 20th century it was widely believed that multistep methods, which, due to the use of several consecutive solution approximations, can be so easily extended to high orders, were in

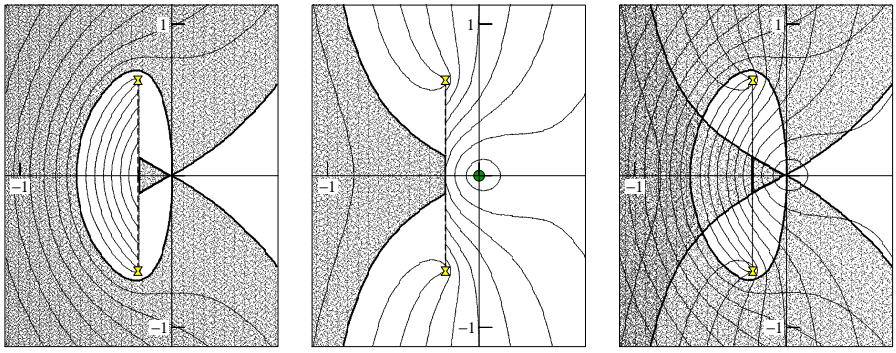

Fig. 4. Order star for Adams2; principal root and parasitic root (left and middle); on Riemann surface (right).

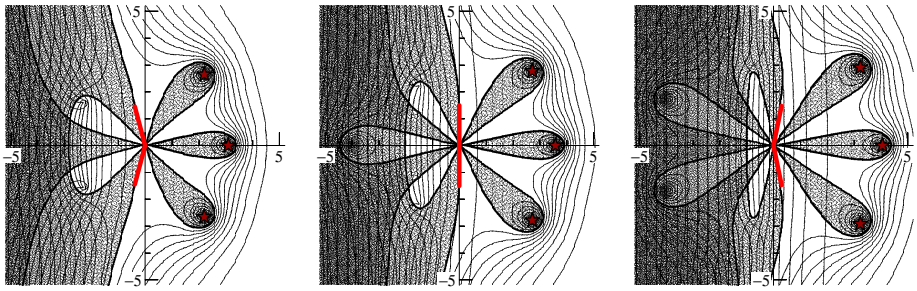

Fig. 5. Two-step method with 3 poles, order 5 (left; A-stable), order 6 (middle; A-stable), order 7 (right; not A-stable); $R(z)$ given by Butcher-Chipman approximations to e^z.

any respect superior to one-step methods. Slowly, during the sixties and seventies, it became apparent that this high order must be paid with lack of stability. Several papers of Jeltsch and Nevanlinna then brought more and more light into this question. Together with order star theory this then led to many important results in [10] and [11]. We illustrate the idea by comparing the stability domain of Runge–Kutta2 with that of Adams2 (see first picture of Fig. 6): RK2 possesses a larger stability domain than Adams2, but RK2 requires two derivative evaluations per step and Adams2 only one. So, for a fair comparison we must *scale* the stability domain, i.e. we replace $R_{\mathrm{RK2}}(z)$ by $\sqrt{R_{\mathrm{RK2}}(2z)}$ (second picture of Fig. 6). We get a nice surprise: *Neither method is always more stable than the other.*

In order to explain this, we compare the stability function of a method, not to $|\mathrm{e}^z|$ as in (2), but to the stability function of *the other* method, i.e. we replace (2) by

$$B := \left\{ z \in \mathbb{C} \; ; \; \frac{|R_1(z)|}{|R_2(z)|} > 1 \right\} . \tag{3}$$

We choose for $R_1(z)$ the pricipal root of Adams2 and for $R_2(z)$ the scaled stability function of RK2 (third picture of Fig. 6). The zeros of $R_2(z)$ turn into poles of the ratio and the order star must cross the scaled stability boundary of RK2 (both methods are explicit and have the same behavior at infinity).

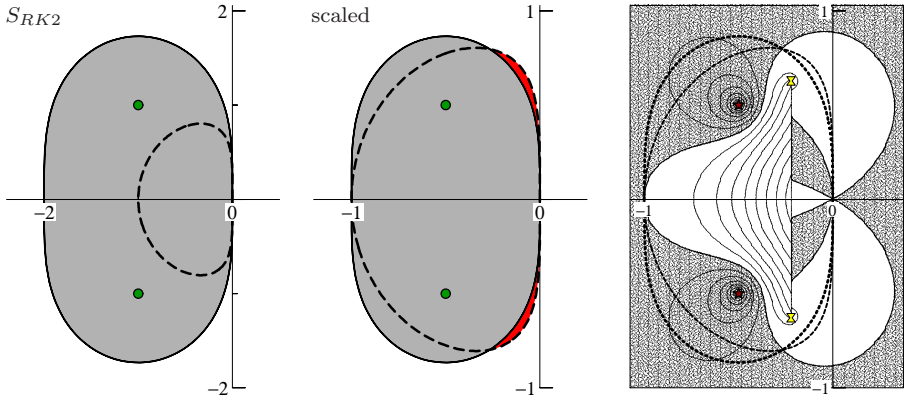

Fig. 6. Comparing stability domains for RK2 vs. Adams2 (left), *scaled* RK2 vs. Adams2 (middle), Proof of the Jeltsch-Nevanlinna theorem (right).

Notes

Further results concern the Nørsett–Wolfbrandt conjecture for approximations with real poles (DIRK and SIRK methods), Abdulle's proof [1] of the Lebedev conjecture for Chebyshev approximations of high order, and an application to delay differential equations [6]. For a thorough treatment of order stars see Sections IV.4 and V.4 of [8] as well as the monograph [9]. A variant of order stars – order arrows – has been introduced by Butcher [2].

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